

Investigation of Effects of Texture Contrast and Background Luminance on Depth Perception in Virtual Reality

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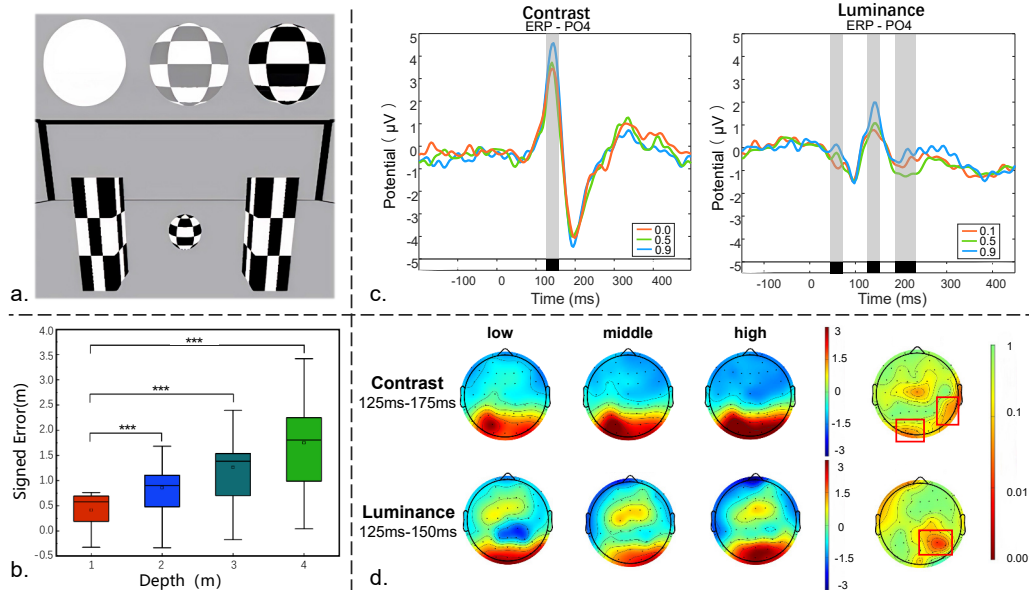


Figure 1: a. VR experimental scene and texture contrast conditions; b. Effects of depth on signed-error performance; c. ERP responses under different contrast and luminance conditions; d. Topographic distribution of significant neural activations.

ABSTRACT

Virtual reality (VR) offers immersive environments. However, it heavily relies on accurate depth perception to support natural interaction and visual comfort. Understanding how visual cues influence this perceptual process is essential for creating stable and realistic VR experiences. This study examines how variations in texture contrast and background luminance affect users' depth judgments and early neural responses. By combining a depth-matching task with electroencephalography (EEG), we assessed whether these cues modulate perceptual sensitivity during early visual processing. Results show that although contrast and luminance significantly influence early neural activity, these effects do not translate into measurable improvements in explicit depth judgments.

Index Terms: virtual reality, depth perception, contrast, EEG.

1 INTRODUCTION

Virtual reality (VR) has been widely used across entertainment, education, healthcare, and industrial applications, yet achieving stable and accurate depth perception remains challenging. In particu-

lar, the mismatch between vergence and accommodation can reduce spatial accuracy and degrade user experience.

In personal and action spaces commonly involved in VR interaction, depth cues such as binocular disparity, motion parallax, relative size, and surface texture are considered the primary factors affecting depth perception [1]. Luminance contrast can enhance depth perception in optical see-through AR, but adding more cues does not always improve accuracy [2].

Electroencephalography (EEG) provides a means to examine the neural basis of visual perception with high temporal precision. Variations in contrast, luminance, and depth have been shown to modulate early visual event-related potential (ERP) components associated with sensory processing. However, how texture contrast and background luminance jointly influence depth perception in immersive VR remains insufficiently understood. This study therefore combines behavioral measures and EEG to investigate how these visual factors contribute to perceptual and neural responses in VR depth judgment.

2 METHOD

To evaluate these effects with both behavioral and neural measures, we adopted the following experimental setup and participant configuration.

2.1 Participants and Apparatus

Twelve right-handed participants (6 males, 6 females; age 20–26, $M = 23$, $SD = 2.19$) took part in the study. All reported normal or corrected-to-normal vision and no neurological disorders. The

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study was approved by the institutional ethics committee, and participants received monetary compensation.

The VR experiment was conducted using an HTC Vive Pro head-mounted display and a Vive Controller 2.0 for user interaction. The virtual environment was developed in Unity. EEG signals were recorded using a 64-channel Neuroscan SynAmps 2 system arranged according to the international 10–20 system, with impedances maintained below 10 k Ω and a sampling rate of 1000 Hz.

2.2 Experimental Design

In VR interaction tasks, the user's manipulation method directly affects perceptual efficiency in estimating spatial attributes of virtual environments. We employed a perceptual matching paradigm, in which participants moved a target sphere along the depth axis to align with left and right reference rectangles. All virtual objects used black-and-white checkerboard textures to minimize perceptual error[2]. The target sphere was initially set to be 0.5 m from the observer, with contrast levels set to 0.0, 0.5, and 1.0. Cubes with a contrast of 1.0 served as the reference objects. Two cubes were placed symmetrically at one of four depths (1 m, 2 m, 3 m, 4 m) relative to the sphere's initial position to eliminate disparity bias. The background luminance was set to three levels (0.1, 0.5, 0.9). A schematic diagram of the VR experimental scene is shown in Fig. 1a.

Luminance followed ITU-R BT.709, with Unity linear color space enabling accurate grayscale-to-luminance mapping. Checkerboard texture contrast was quantified using Michelson contrast.

2.3 Procedure

During the experiment participants wore an EEG cap with electrode impedances maintained below 10 k Ω throughout the experiment. After system calibration and brief practice trials, the formal experiment began. In each trial, participants adjusted the target position along the depth axis using a handheld controller and confirmed their judgment by pressing a trigger button.

All experimental conditions were repeated 15 times following a Latin square design to counterbalance order effects. After the experiment, participants completed a brief 7-point post-questionnaire on fatigue and overall experience and participated in a short interview.

2.4 Data Analysis

Both VR behavioral data and EEG signals were synchronized via timestamps. For behavioral analysis, the signed error was calculated as the difference between the matched and target depth. Data normality was tested using the Shapiro–Wilk test; non-normal data ($p < 0.05$) justified the use of a Generalized Linear Mixed Model (GLMM), with participant as a random effect and cube depth, sphere contrast, and luminance as fixed factors.

EEG data were processed in EEGLAB: resampled to 256 Hz, filtered (0.5–40 Hz), and cleaned using cleanLineNoise and ASR algorithms, which attenuate line-noise harmonics and automatically detect/correct high-variance artifact segments, respectively. Independent Component Analysis (AMICA) and ICLabel were used to remove artifacts. Epochs ranged from -0.8 to 1.5 s relative to stimulus onset, with a baseline of -190 to 0 ms. ERPs were averaged across identical luminance and contrast conditions for group-level analysis.

3 RESULTS

GLMM results showed a significant main effect of depth on signed error ($p < 0.001$). Compared to the 1 m baseline, signed error increased by 0.45 m at 2 m, 0.85 m at 3 m, and 1.38 m at 4 m, indicating systematic overestimation with increasing distance

(Fig. 1b). The main effects of sphere contrast and background luminance were not significant ($p > 0.05$). Interaction effects (including higher-order interactions) did not reach significance ($p > 0.05$) either. Questionnaire results indicated mild eye strain or fatigue in some participants ($M = 3.92$, $SD = 1.44$). Post-experiment interviews suggested that relative size had a stronger influence on alignment judgments than surface texture.

As shown in Fig. 1c, significant modulation by contrast and luminance was observed at the parieto-occipital site PO4, where the effects were strongest. Under contrast conditions, N1 amplitude increased with contrast during the -100 – 100 ms and 125 – 175 ms time windows, with high-contrast stimuli eliciting sharper negative peaks, indicating enhanced visual attention and depth cue extraction. Under luminance conditions, P2 amplitude increased with luminance during the 120 – 150 ms time window, suggesting facilitated early-stage depth information processing. Topographical maps showed that neural activity was mainly concentrated in the right parietal–occipital region (Fig. 1d), associated with spatial and depth processing.

In summary, these findings indicate that texture contrast and background luminance jointly enhance early sensory encoding at the neural level, with contrast primarily strengthening feature encoding and luminance facilitating subsequent depth-related processing. However, despite this clear modulation of early visual ERP components (N1/P2), these neural enhancements did not lead to improved explicit depth-matching performance. This dissociation suggests that enhanced early sensory encoding alone may be insufficient to support more accurate depth judgments in the current task. Instead, tasks with finer spatial resolution or alternative behavioral paradigms (e.g., forced-choice or JND-based measures) may be required to reveal whether such neural enhancements can be behaviorally leveraged. Consistent with prior work[3], this EEG–behavior dissociation likely reflects a perceptual–cognitive separation: contrast and luminance modulated early sensory processing, but the scale of the current task—particularly object size and movement granularity—limited the translation of enhanced sensory encoding into observable behavioral gains.

4 CONCLUSION

This study combined EEG and behavioral measures to examine the effects of texture contrast and background luminance on depth perception in VR. Although both cues significantly influenced early neural processing, these effects did not translate into improved behavioral depth judgments. This dissociation suggests that the current task may have lacked sufficient sensitivity to capture subtle perceptual enhancements. Future studies employing finer spatial resolution and more sensitive behavioral paradigms may help determine whether these enhancements in early sensory processing can be translated into more accurate depth judgments in VR.

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